Lihua Lei

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Academic Appointments

Assistant Professor of Economics — 7/2022 - present
Stanford Graduate School of Business

Assistant Professor of Statistics (by courtesy) — 11/2022 - present
Department of Statistics

Faculty Fellow — 12/2022 - present
Stanford Institute for Economic Policy Research

Postdoctoral Researcher, Stanford University — 9/2019 - 7/2022
Advisor: Emmanuel Candès

Education

Ph.D. Statistics, University of California, Berkeley — 8/2014 - 8/2019
Advisors: Peter, J. Bickel, and Michael, I. Jordan

B.S. Statistics, Peking University — 9/2010 - 6/2014
Undergraduate Thesis Advisor: Song Xi, Chen

B.A. Economics (double major), Peking University — 9/2011 - 6/2014

Research

Journal Publications [1]

1. Sahoo, R., **Lei, L.**, and Wager, S. (2022) Learning from a Biased Sample. *Arxiv e-prints, abs/2209.01754*. Accepted at *Management Science (MS)*.
2. Luo, Y., Fithian, W. and **Lei, L.** (2026) Estimating the FDR of variable selection. *Arxiv e-prints, abs/2408.07231*. Accepted at *Annals of Statistics (AoS)*.
3. Luo, Y., Fithian, W, and **Lei, L.** (2025) Improving knockoffs with conditional calibration. *Arxiv e-prints, abs/2208.09542*. Accepted at *Annals of Statistics (AoS)*.
4. Angelopoulos, A.*, Bates, S.*, Candès, E. J.*, Jordan, M. I.*, and **Lei, L.*** (2025) Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. *Annals of Applied Statistics (AoAS)*, 19(2), 1641-1662

[1] * = alphabetical ordering or equal contribution

5. **Lei, L.*** (2024) Discussion of "Data Fission: Splitting a Single Data Point" – Some Asymptotic Results for Data Fission. *Journal of the American Statistical Association (JASA), 120(549), 147-150,* (invited discussion paper).

6. Arkhangelsky, D., Imbens, G., **Lei, L.**, and Luo, X. (2024) Design-Robust Two-Way-Fixed-Effects Regression For Panel Data. *Quantitative Economics (QE)*, 15(4), 999-1034.

7. Marandon, A., **Lei, L.**, Mary, D., and Roquain, R. (2024) Adaptive novelty detection with false discovery rate guarantee. *Annals of Statistics (AoS)*, 52(1), 157-183.

8. Spector, A., Candès, E. J.*, **Lei, L.*** (2023) A Discussion of Tse and Davidson (2022) "A Note on Universal Inference". *Stat*, 12(1)

9. Bates, S.*, Candès, E. J.*, **Lei, L.***, Romano, Y.*, and Sesia, M.* (2022) Testing for Outliers with Conformal p-values. *ArXiv e-prints, abs/2104.08279. Annals of Statistics (AoS)*, 51(1), 149-178.

10. Candès, E. J.*, **Lei, L.***, Ren, Z.* (2022) Conformalized Survival Analysis. *ArXiv e-prints, abs/2103.09763. Journal of the Royal Statistical Society: Series B (JRSS-B)*, 85(1), 24-45.

11. Horváth, S.*, **Lei, L.***, Richtárik, P., and Jordan, M. I. (2022) Adaptivity of Stochastic Gradient Methods for Nonconvex Optimization. *SIAM Journal on Mathematics of Data Science (SIMODS)*, 4(2), 634-648.

12. Fithian, W. and **Lei, L.** (2022) Conditional calibration for false discovery rate control under dependence. *Annals of Statistics (AoS)*, 50(6), 3091-3118.

13. Bates, S.*, Angelopoulos, A.*, **Lei, L.***, Malik, J., and Jordan, M. I. (2021) Distribution-Free, Risk-Controlling Prediction Sets. *Journal of the ACM (JACM)*, 68(6), 1–34.

14. Loper, J. H.*, **Lei, L.***, Fithian, W., and Tansey, W. (2021) Smoothed Nested Testing on Directed Acyclic Graphs. *Biometrika*, 109(2), 457-471.

15. **Lei, L.** and Candès, E. J. (2021) Conformal Inference of Counterfactuals and Individual Treatment Effects. *Journal of the Royal Statistical Society: Series B (JRSS-B)*. 83(5), 911-938.

16. **Lei, L.** and Ding, P. (2020) Regression Adjustment in Randomized Experiments With A Diverging Number of Covariates. *Biometrika*. 108(4), 815-828.

17. Li, T., **Lei, L.**, Bhattacharyya, S., Van den Berge, K., Sarkar, P., Bickel, P. J., and Levina, E. (2020) Hierarchical community detection by recursive partitioning. *Journal of the American Statistical Association (JASA)* 117(538), 951-968.

18. **Lei, L.** and Bickel, P. J. (2020) An Assumption-Free Exact Test For Fixed-Design Linear Models With Exchangeable Errors. *Biometrika*, 108(2), 397-412.

19. **Lei, L.**, Ramdas, A., and Fithian, W. (2020) A general interactive framework for FDR control under structural constraints. *Biometrika*, 108(2), 253-267.

20. **Lei, L.** and Jordan, M. I. (2020) On the Adaptivity of Stochastic Gradient-Based Optimization. *SIAM Journal on Optimization (SIOPT)*, 30(2), 1473-1500.

21. D'Amour, A., Ding, P., Feller, A., **Lei, L.**, and Sekhon, J. (2019) Overlap in High Dimensional Observational Studies. *Journal of Econometrics (JoE)*, 221(2), 644-654.
22. **Lei, L.**, and Fithian, W. (2018). AdaPT: An Interactive Procedure For Multiple Testing With Side Information. *Journal of the Royal Statistical Society: Series B (JRSS-B)*, 80(4), 649-679.
23. **Lei, L.**, Bickel, P. J., and El Karoui, N. (2018). Asymptotics For High Dimensional Regression M-Estimates: Fixed Design Results. *Probability Theory and Related Fields (PTRF)*, 172(3-4), 983-1079.
24. Chen, S. X., **Lei, L.**, and Tu, Y. (2016). Functional Coefficient Moving Average Model with Applications to forecasting Chinese CPI. *Statistica Sinica*, 26, 1649-1672.

Conference Publications

1. Chen, Y.*, **Lei, L.** (2026) Compound Estimation for Binomials. Accepted at *The 27th ACM Conference on Economics and Computation (EC).*
2. Chen, C., Shen, J., Deng, Z., **Lei, L.** (2026) Adversarially Robust Control of Conditional Value-at-Risk via Kelly Conformal Inference. Accepted at *The 43rd International Conference on Machine Learning (ICML).*
3. Li, X., Li, Y., Zhong, H., **Lei, L.** and Deng, Z. (2025). Statistical Inference under Performativity. arXiv preprint. In *The 39th of Annual Conference on Neural Information Processing Systems (NeurIPS).*
4. Andrews, I., Fudenberg, D., **Lei, L.**, Liang A., and Wu. C. (2023) The Transfer Performance of Economics Models. Accepted at *The 26th ACM Conference on Economics and Computation (EC).*
5. Chen, C., Shen, J., Deng, Z., **Lei, L.** (2025) Conformal Tail Risk Control for Large Language Model Alignment. In *Proceedings of the 42th International Conference on Machine Learning (ICML).*
6. Wang, W., Janson, L., **Lei, L.**, and Ramdas, A. (2024) Total Variation Floodgate for Variable Importance Inference in Classification. In *Proceedings of the 41th International Conference on Machine Learning (ICML).*
7. Angelopoulos, A.*, Bates, S.*, Fisch, A.*, **Lei, L.***, and Schuster T.* (2023) Conformal Risk Control. In *International Conference on Learning Representations (ICLR).*
8. Elibol, M., **Lei, L.**, and Jordan, M. I. (2020). Variance Reduction with Sparse Gradients. In *International Conference on Learning Representations (ICLR).*
9. Ye, Y., **Lei, L.**, and Ju, C. (2018). HONES: A Fast and Tuning-free Homotopy Method For Online Newton Step. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS).*
10. **Lei, L.**, Ju, C., Chen, J., and Jordan, M. I. (2017). Nonconvex Finite-Sum Optimization Via SCSG Methods. In *Proceedings of the 30th Advances in Neural Information Processing Systems (Neurips).*

11. **Lei, L.** and Jordan, M. I. (2017). Less than a Single Pass: Stochastically Controlled Stochastic Gradient. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS).*

12. **Lei, L.** and Fithian, W. (2016). Power of Ordered Hypothesis Testing. In *Proceedings of the 33th International Conference on Machine Learning (ICML).*

Under Revision

1. **Lei, L.*** and Ross, B.* (2023) Estimating Counterfactual Matrix Means with Short Panel Data. *Arxiv e-prints, abs/2312.07520.* Under Revise and Resubmit at *Econometrica*.

2. Ji, W.*, **Lei, L.***, and Spector A.* (2023) Model-Agnostic Covariate-Assisted Inference on Partially Identified Causal Effects. *Arxiv e-prints, abs/2310.08115.* Under Major Revision at *Journal of Political Economy (JPE).*

3. Ji, W.*, **Lei, L.***, and Zrnic, T.* (2025) Predictions as surrogates: Revisiting surrogate outcomes in the age of AI. *ArXiv preprint abs/2501.09731.* Under Major Revision at *Biometrika*.

4. Hartog, W.*, **Lei, L.*** (2025) Family-wise error rate control with e-values. *ArXiv preprint abs/2501.09015.* Under Major Revision at *Annals of Statistics (AoS).*

Preprints and Submissions

1. **Lei, L.*** (2026) How Much Can Gaussian Dependence Inflate the Benjamini–Hochberg Procedure's FDR? *ArXiv e-prints abs/2607.14812.*

2. Chen, C.*, Gao, C.*, Hazell, J.*, **Lei, L.***, and Lian, C.* (2026). Forecasting Inflation with Microdata: An Adaptive Machine Learning Approach. *ArXiv e-prints abs/2607.12345.*

3. Sudijono, T., **Lei, L.**, Masoero, L., Vijaykumar, S., Imbens, G., and McQueen, J. (2026). Regression Adjustments for Double Randomization in Two-Sided Marketplaces. *ArXiv e-prints, abs/2603.19480.*

4. Tan, L., Sagan, N., **Lei, L.**, and Blanchet, J. (2026). When Should Humans Step In? Optimal Human Dispatching in AI-Assisted Decisions. *ArXiv e-prints, abs/2603.13688.*

5. Borusyak, K.*, Chen, J.*, Hull, P.*, **Lei, L.*** (2025) Nonparametric Identification of Demand without Exogenous Product Characteristics. *Arxiv e-prints, abs/2512.23211.*

6. Chen, J.*, **Lei, L.***, Sudijono, T*., Sun, L.*, and Xie, T.* (2025) Compound Selection Decisions: An Almost SURE Approach. *Arxiv e-prints, abs/2511.11862.*

7. Ji, W., Pan, Y., Zhu, R. and **Lei, L.** (2025) Multi-Armed Bandits With Machine Learning-Generated Surrogate Rewards. *Arxiv e-prints, abs/2506.16658.*

8. **Lei, L.** (2024) Causal Interpretation of Regressions With Ranks. *Arxiv e-prints, abs/2406.05548.*

9. Yang, Z., **Lei, L.**, and Candès, E. J. (2024) Bellman Conformal Inference: Calibrating Prediction Intervals For Time Series. *Arxiv e-prints, abs/2402.05203.*

10. **Lei, L.*** and Sudijono, T.* (2024) Inference for Synthetic Controls via Refined Placebo Tests. *Arxiv e-prints, abs/2401.07152.*

11. Viviano, D., **Lei, L.**, Imbens, G., Karrer, B., Schrijvers, O., and Shi, L. (2023) Causal clustering: design of cluster experiments under network interference. *Arxiv e-prints, abs/2310.14983.*

12. **Lei, L.***, Sahoo, R.*, and Wager, S.* (2023) Policy Learning under Biased Sample Selection. *Arxiv e-prints, abs/2304.11735.*

13. **Lei, L.** and Wooldridge, J. (2022) What Estimators Are Unbiased For Linear Models? *Arxiv e-prints, abs/2212.14185.*

14. Yang, C., **Lei, L.**, Ho, N., and Fithian, W. (2021) BONuS: Multiple multivariate testing with a data-adaptive test statistic. *ArXiv e-prints, abs/2106.15743.*

15. **Lei, L.***, Li, X.*, and Lou, X.* (2020) Consistency of Spectral Clustering on Hierarchical Stochastic Block Models. *ArXiv e-prints, abs/2004.14531.*

16. **Lei, L.** (2019) Unified $\ell_{2\to\infty}$ Eigenspace Perturbation Theory for Symmetric Random Matrices. *ArXiv e-prints, abs/1909.04798.*

Software

1. `adaptMT`: `R` package on Adaptive P-value Thresholding (on CRAN);

2. `cfcausal`: `R` package on conformal inference of counterfactuals and individual treatment effects (on github);

3. `dbh`: `R` package on dependence-adjusted Benjamini-Hochberg and general step-up procedures (on github);

4. `mkn`: `R` package on multiple knockoffs based inference (on github).

5. `ovalue`: `R` package on distribution-free assessment of population overlap (O-values) for observational studies (on github).

6. `transferUQ`: `R` package on uncertainty quantification of transfer performance for prediction models

Awards

ICSA Outstanding Young Reseacher Award, 2025.

Bernoulli Society New Reseacher Award, 2025.

NSF CAREER award, 2024.

Fletcher Jones Faculty Scholar for 2023-24.

ICSA Junior Research Award, 2022.

IMS (Institute of Mathematical Statistics) New Researcher Travel Award, 2022.

Rising Star in Data Science (University of Chicago), 2021.

Eric Lehmann Citation (Annual Dissertation Award in Theoretical Statistics), 2019.

Citadel Fellowship, 2017-2018.

Outstanding Graduate Student Instructor Award, 2016.

ICML (International Conference on Machine Learning) Travel Award, 2016.

ACIC (Atlantic Causal Inference Conference) Travel Award, 2018, 2022.

Scholarship from The Sally and Terry Speed Graduate Support Fund, 2015.

Invited Talks

2026: Cornell (Econ), UCSD (Econ), Stanford (Stats), Cowles Foundation Summer Conference in Econometrics (Yale), NBER Summer Institute (Forecasting & Empirical Methods), Joint Statistical Meetings (JSM), EcoSta Conference, Southern Economic Association (SEA) Annual Meeting.

2025: Joint Statistical Meetings (JSM), Workshop on "Econometrics at Emory: Causal Inference with Panel Data" (Emory), Princeton (Econ), Berkeley (Econ), Duke (Decision Sciences), Southern Economic Association (SEA) Annual Meeting, Washington University in St. Louis (Stats).

2024: UBC (Marketing, Stats, Econ), SFU (Econ), NYU (Econ), UCLA (Econ), NorCal Junior Econometricians' Conference (UCSC), Cowles Foundation Summer Conference in Econometrics (Yale), Workshop of Statistical Network Analysis and Beyond (SNAB 2024), Pacific Causal Inference Conference (PCIC), EcoSta Conference, Joint Conference on Statistics and Data Science (JCSDS), 2024 Shanghai Workshop on Robustness Meets Causality, NBER Summer Institute (Labor Studies), Upenn (Econ, Stats), PennState (Econ), MIT (IDSS), David Simchi-Levi's group (MIT), Wisconsin (Econ), UW (Econ), Southern Economic Association (SEA) Annual Meeting, International Conference on Statistics and Data Science (ICSDS).

2023: McGill (Stats), UNC Kosorok PHAIR Lab, Cornell (Stats), University of Warwick (Guest lecture), Econometrics in the Era of Machine Learning (University of Chicago), Columbia (IEOR and DRO), Rutgers (Econ), Peking University (Stats), Clubear, Brown (Econ), NorCal Junior Econometricians' Conference (UC Davis), Workshop of Statistical Network Analysis and Beyond (SNAB 2023), Yale (Econ), Columbia (Econ), Northwestern (Econ), UChicago (Econ), Southern Economic Association (SEA) Annual Meeting.

2022: Chamberlain Econometrics Seminar, Online Causal Inference Seminar, UC Berkeley (Econ), Joint Statistical Meetings, ICSA China Conference, Workshop on Interactive Causal Learning, IMS Annual in Probability and Statistics, Simons Institute for the Theory of Computing (UC Berkeley), New England Statistics Symposium, Tsinghua IIIS Seminar on Foundations Of Data Science, UNC CIRG Seminar, AEA/ASSA Annual Meeting

2021-22 job talks: University of Michigan (Stats), USC (Math), Rutgers (Stats), NYU Courant (Data Science), USC(Stats), University of Minnesota (Stats), Northwestern (Stats), Stanford GSB (Econ), Columbia (Stats), UPenn Wharton (Stats), CMU (Stats), Harvard (Stats), University of Chicago

(Data Science), Chicago Booth (Econometrics), Purdue (Stats), UC Berkeley (Stats), Yale (Stats)

2021: One World YoungStatS Webinar, Journal of Royal Statistical Society Webinar, Stanford Causal Inference Group, University of Washington (Stats), UC Davis (Econ), Harvard (Econ), Temple University (Stats), NYU Data Science Lunch Seminar, ICSA Applied Statistics Symposium, International Conference on Multiple Comparison Procedures, ICML Workshop on "Neglected Assumptions of Causal Infernece", IFDS Summer Workshop, INFORMS Virtual Healthcare Conference, The Hong Kong University of Science & Technology (Stats), WNAR Annual Meeting, ETH Zürich (Data Science), UC Berkeley (Stats), Stanford (Stats), University of Chicago (Stats), Stanford RAIN Seminar, UChicago Crime Lab, Chicago Booth (Econometrics), University of Maryland, Baltimore County (Stats), Dataiku Research, Northwestern (Biostats, Stats), University of Manchester (Biostats), The London School of Hygiene & Tropical Medicine, Computer Vision Talks Webinar, Facebook Core Data Science, UCLA (Big Data and ML), University of Missouri (Biostats), Leiden Univerity, Twitter, Stitch Fix, Rutgers (Stats), UPenn Wharton (Stats)

2020: Lyft, John Hopkins University Causal Inference Group, Florida State University (Guest Lecture), UC Berkeley (Guest Lecture), University of North Carolina at Chapel Hill (Biostats), Peking University (Biostats), Pacific Causal Inference Conference, Online Causal Inference Seminar, International Seminar on Selective Inference, Southwestern University of Finance and Economics, BAAI Conference, Stanford University (Journal Club), Stanford Data Studio, UC Berkeley (Guest Lecture), WNAR Annual Meeting, UC Berkeley Causal Inference Group

2019: SAMSI Workshop, Facebook Core Data Science, Berkeley-Stanford Econometrics Jamboree, Joint Statistical Meetings, ICSA Applied Statistics Symposium, Yale (Polisci), UC Davis (Stats), Stanford University (Guest Lecture)

2018: USC (Stats), University of Michigan (Guest Lecture), University of Michigan (Stats), International Conference on Econometrics and Statistics, Atlantic Causal Inference Conference, UC Davis (Guest Lecture), Stanford University (Guest Lecture)

2017: Berkeley-Stanford Econometrics Jamboree

Reviewing[2]

Statistics Journals (90): (#papers in parentheses) Annals of Statistics (AoS, 20), Journal of the American Statistical Association (JASA, 17), Biometrika (13), Journal of the Royal Statistical Society-Series B (JRSS-B, 12), Journal of the Royal Statistical Society-Series A (JRSS-A, 1), Bernoulli (2), Electronic Journal of Statistics (EJS, 5), Biometrics (3), Journal of Causal Inference (JCL, 2), Stat (1), Springer Book (1), SpringerBriefs (1), International Journal of Biostatistics (IJB, 1), Journal of Computational and Graphical Statistics (JCGS, 2), Statistica Sinica (3), International Journal of Approximate Reasoning (IJA, 1), Statistics in Biosciences (SIBS, 1), Journal of Statistical Planning and Inference (JSPI, 2), Statistical Science (SS, 2), Statistics (1)

Economics Journals (34): (#papers in parentheses) Econometrica (6), Journal of Political Economy (JPE, 6), The Quarterly Journal of Economics (QJE, 1), Review of Economic Studies (ReStud, 2),

[2] reviews of revisions not included

JPE Mirco (3), The Review of Economics and Statistics (ReStat, 7), Journal of Applied Econometrics (JAE, 1), Journal of Business and Economic Statistics (JBES, 2), Journal of Econometrics (JoE, 6)

Other Journals (26): Proceedings of the National Academy of Sciences of the United States of America (PNAS, 2), SIAM Journal on Mathematics of Data Science (SIMODS, 1), Operations Research (1), Management Science (1), Journal of Machine Learning Research (JMLR, 9), Transactions on Pattern Analysis and Machine Intelligence (TPAMI, 3), Computational Optimization and Applications (COAP, 1), Optimization Methods and Software (GOMS, 1), Artificial Intelligence (1), Information and Inference: A Journal of the IMA (IMAIAI, 2), Vietnam Journal of Mathematics (VJOM, 1), IEEE Transactions on Information Theory (1), American Journal of Epidemiology (AJE, 1), BMC Medical Research Methodology (1)

Conferences: (year in parentheses) ICML (2019, 2020, 2021, 2026), NeurlPS (2019, 2020, 2021), COLT (2019, 2020, 2021), AISTATS (2019), UAI (2020, 2021)

Other Academic Services

- Co-organizer of the International Seminar on Selective Inference (with Will Fithian, Rina Barber, and Jelle Goeman, 2020-present).
- Co-organizer of the Causal Panel Data Conference (with Bryan Graham, Guido Imbens, and Jann Spiess, 2023).
- Institute of Mathematical Statistics (IMS) Committee on Nomations.
- Co-organizer of the GRoup-Of-Why (GROW) seminar, a causal reading group at Stanford University (with Guillaume Basse and Dominik Rothenhäusler, 2019).
- Invited panelist at "IMS NRG Virtual Panel: Navigating the Academic Job Market" (2024)